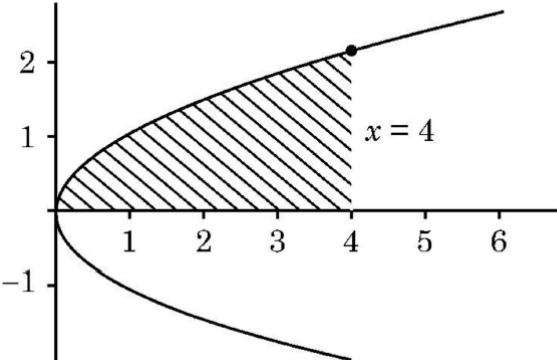
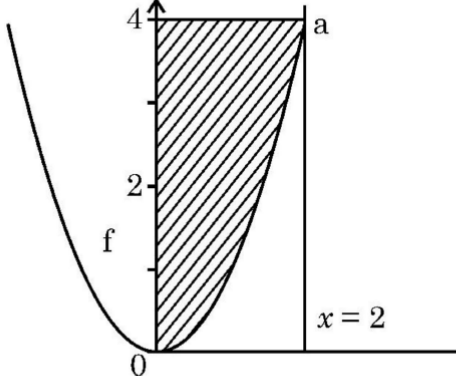


APPLICATION OF INTEGRALS

1.	Find the area of the region bounded by the curves $x^2 = y$, $y = x + 2$ and x-axis, using integration.
2.	Using Integration, find the area of triangle whose vertices are $(-1, 1)$, $(0, 5)$ and $(3, 2)$.
3.	The area of the region bounded by the line $y = mx$ ($m > 0$), the curve $x^2 + y^2 = 4$ and the x-axis in the first quadrant is $\frac{\pi}{2}$ units. Using integration, find the value of m .
4.	Using integration, find the area of the region bounded by the parabola $y^2 = 4ax$ and its latus rectum.
5.	Sketch the region bounded by the lines $2x + y = 8$, $y = 2$, $y = 4$ and the y-axis. Hence, obtain its area using integration.
6.	Find the area of the minor segment of the circle $x^2 + y^2 = 4$ cut off by the line $x = 1$, using integration.
7.	The area (in sq. units) of the region bounded by the curve $y = x$, x-axis, $x = 0$ and $x = 2$ is : (A) $\frac{3}{2}$ (B) $\frac{1}{2} \log 2$ (C) 2 (D) 4
8.	Using integration, find the area of the region bounded by the curve $y = \sqrt{4 - x^2}$, the lines $x = -\sqrt{2}$ and $x = \sqrt{3}$ and the x-axis.
9.	Using integration, evaluate the area of the region bounded by the curve $y = x^2$, the lines $y = 1$ and $y = 3$ and the y-axis.
10.	Find the area of the triangle ABC bounded by the lines represented by the equations $5x - 2y - 10 = 0$, $x - y - 9 = 0$ and $3x - 4y - 6 = 0$, using integration method.
11.	Using integration, find the area of the ellipse $\frac{x^2}{16} + \frac{y^2}{4} = 1$, included between the lines $x = -2$ and $x = 2$.

12.	If A_1 denotes the area of region bounded by $y^2 = 4x$, $x = 1$ and x -axis in the first quadrant and A_2 denotes the area of region bounded by $y^2 = 4x$, $x = 4$, find $A_1 : A_2$.
13.	Using integration, find the area of the region enclosed between the curve $y = \sqrt{4-x^2}$ and the lines $x = -1$, $x = 1$ and the x -axis.
14.	Sketch the graph of $y = x x $ and hence find the area bounded by this curve, X -axis and the ordinates $x = -2$ and $x = 2$, using integration.
15.	Using integration, find the area bounded by the ellipse $9x^2 + 25y^2 = 225$, the lines $x = -2$, $x = 2$, and the X -axis.
16.	Area of the region bounded by curve $y^2 = 4x$ and the X -axis between $x = 0$ and $x = 1$ is : (A) $\frac{2}{3}$ (B) $\frac{8}{3}$ (C) 3 (D) $\frac{4}{3}$
17.	Find the area of the region bounded by the curve $4x^2 + y^2 = 36$ using integration.
18.	The area of the shaded region bounded by the curves $y^2 = x$, $x = 4$ and the x-axis is given by  (A) $\int_0^4 x \, dx$ (B) $\int_0^2 y^2 \, dy$ (C) $2 \int_0^4 \sqrt{x} \, dx$ (D) $\int_0^4 \sqrt{x} \, dx$
19.	Sketch the graph of $y = x + 3 $ and find the area of the region enclosed by the curve, x -axis, between $x = -6$ and $x = 0$, using integration.

20.	The area of the shaded region (figure) represented by the curves $y = x^2$, $0 \leq x \leq 2$ and y-axis is given by  (A) $\int_0^2 x^2 dx$ (C) $\int_0^4 x^2 dx$ (B) $\int_0^2 \sqrt{y} dy$ (D) $\int_0^4 \sqrt{y} dy$
21.	Using integration, find the area of the region bounded by the line $y = 5x + 2$, the x – axis and the ordinates $x = -2$ and $x = 2$.
22.	The area of the region enclosed by the curve $y = \sqrt{x}$ and the lines $x = 0$ and $x = 4$ and x-axis is : (A) $\frac{16}{9}$ sq. units (C) $\frac{16}{3}$ sq. units (B) $\frac{32}{9}$ sq. units (D) $\frac{32}{3}$ sq. units
23.	Calculate the area of the region bounded by the curve $\frac{x^2}{9} + \frac{y^2}{4} = 1$ and the x-axis using integration.
24.	The area of the region enclosed between the curve $y = x x$, x-axis, $x = -2$ and $x = 2$ is : (A) $\frac{8}{3}$ (C) 0 (B) $\frac{16}{3}$ (D) 8
25.	Sketch a graph of $y = x^2$. Using integration, find the area of the region bounded by $y = 9$, $x = 0$ and $y = x^2$.

26.	The area of the region bounded by the curve $y^2 = x$ between $x = 0$ and $x = 1$ is : (A) $\frac{3}{2}$ sq units (B) $\frac{2}{3}$ sq units (C) 3 sq units (D) $\frac{4}{3}$ sq units
27.	Use integration to find the area of the region enclosed by curve $y = -x^2$ and the straight lines $x = -3$, $x = 2$ and $y = 0$. Sketch a rough figure to illustrate the bounded region.
28.	A woman discovered a scratch along a straight line on a circular table top of radius 8 cm. She divided the table top into 4 equal quadrants and discovered the scratch passing through the origin inclined at an angle $\frac{\pi}{4}$ anticlockwise along the positive direction of x-axis. Find the area of the region enclosed by the x-axis, the scratch and the circular table top in the first quadrant, using integration.